DAY 01 OF 16 · 3D & 6DOF STREAMING

What is 3D? Why 3D?

From flat pictures to worlds you can walk into: the motivation for the whole course, and the 6 numbers that describe where you are.

Dimensions | 2D vs 2.5D vs 3D | Geometry · topology · semantics | 3DoF vs 6DoF | Lab: see it yourself

Book: Ch. 1 · Code: day01_why_3d/

THE BIG IDEA

A photo tells you what something looks like. 3D tells you where everything is.

Every session in this course builds on that difference, all the way to streaming worlds people can walk through.

BOOK CH. 1 · DIMENSIONS

A dimension is one number you need to describe something

1D

One number per item: the temperature of water, the height along one line.

01

2D

Two numbers: x and y. A photo is a grid of pixels addressed by (x, y).

02

2.5D

A depth map: one distance per pixel, from ONE viewpoint. It cannot see behind things.

03

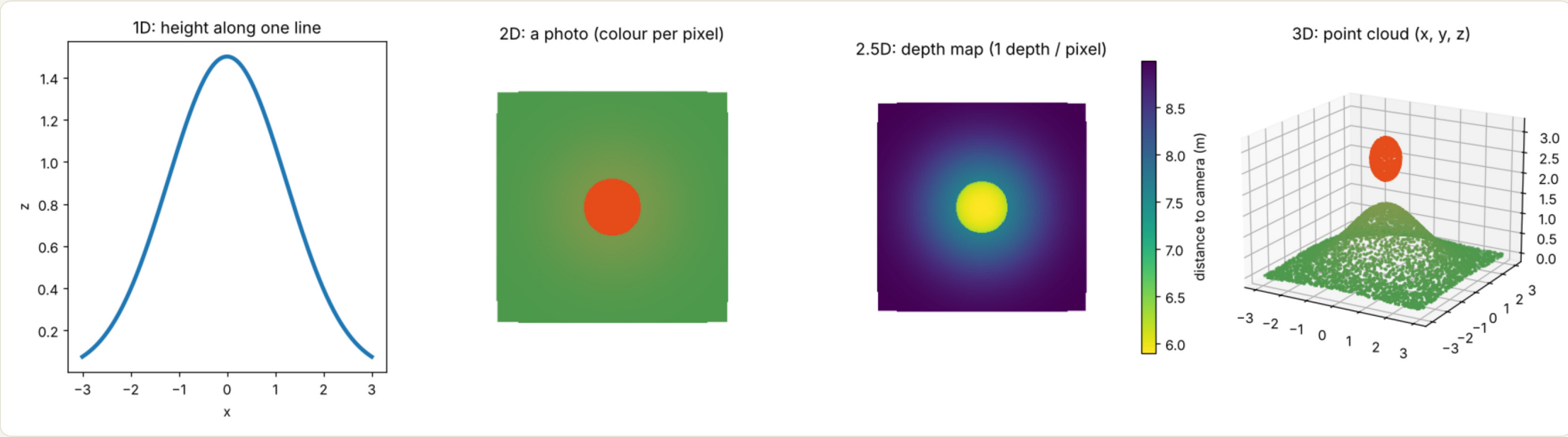
3D

Three numbers per point: x, y, z. Any point, from any side, as a free set.

04

LAB PREVIEW

The same object, four ways



Produced by 01_dimensions.py. In the depth map the hill under the ball is simply gone: one depth per pixel.

BOOK CH. 1 · BUILDING BLOCKS

Three layers turn points into understanding

Geometry

Size, position, orientation, shape. Where is the wall and how tall is it?

01

Topology

How parts connect: the column supports the floor, doors connect rooms.

02

Semantics

What things ARE: this is a kitchen, that is a chair leg.

03

BOOK CH. 1 · POINT CLOUDS

The point cloud: the closest link to reality

What it is

A set of {X, Y, Z} points plus attributes such as colour, intensity or class.

01

Where it comes from

LiDAR, photogrammetry, depth cameras, and now generative AI.

02

Why it is hard

Volume, velocity, variety, veracity, value: the five Vs of big data.

03

THE STREAMING SIDE

Degrees of freedom: how many ways can the viewer move?

3DoF

Rotation only: yaw (turn), pitch (nod), roll (tilt). 360° video. You can look around, but not move.

01

3DoF+

Rotation plus small head movement while seated. Some parallax, limited range.

02

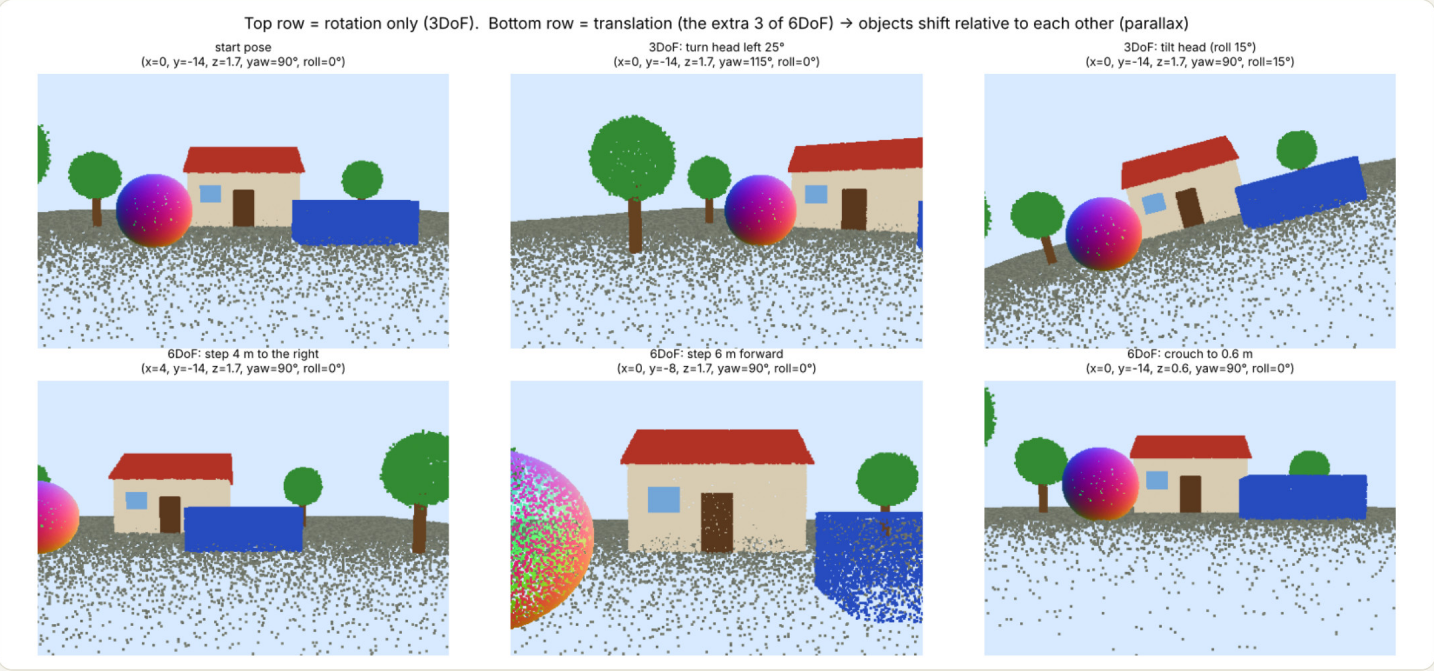
6DoF

Rotation AND translation: x, y, z. Walk, crouch, look behind objects. Needs real 3D content.

03

LAB PREVIEW

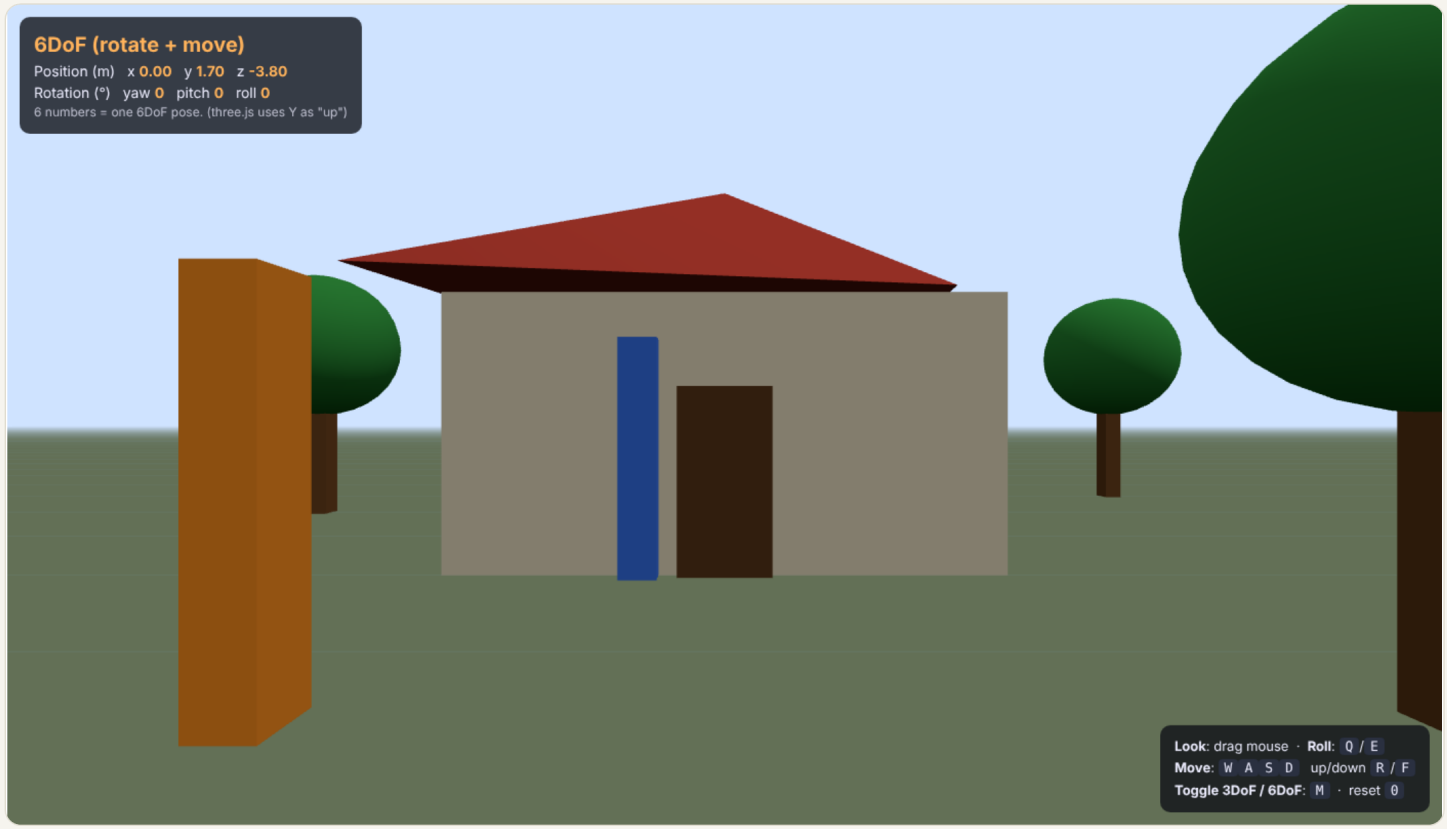
Only translation creates parallax



Top row: rotation only, objects keep their arrangement. Bottom row: moving changes which object is in front of which.

6 NUMBERS

A pose = where you are + where you look



- x, y, z: position in metres
- yaw, pitch, roll: orientation in degrees
- 6DoF client = sends this pose every frame
- Mind the axes: three.js is Y-up, our Python is Z-up

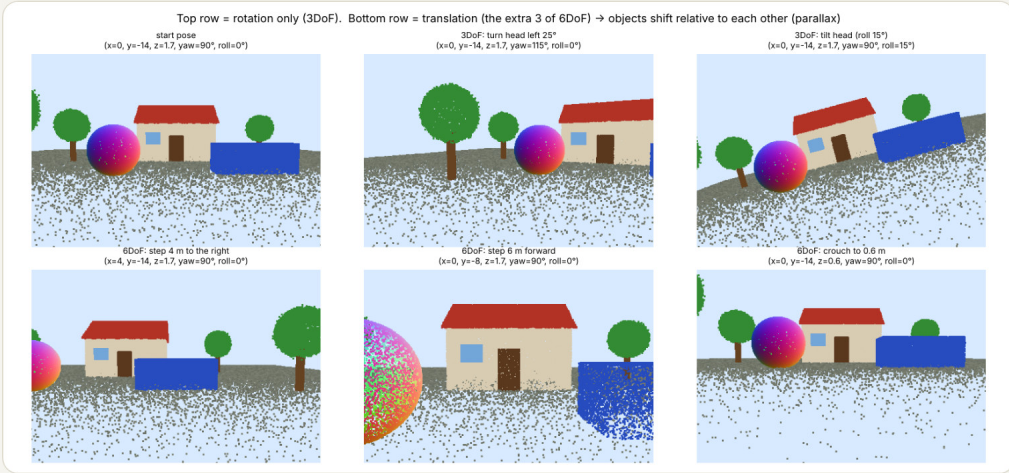
dof_playground.html. Press M to lock translation (3DoF) and try to look behind the orange pillars.

HANDS-ON LAB · COPY, RUN, LOOK

Run your first two scripts and the playground

RUN IT

```
cd day01_why_3d
python 01_dimensions.py
python 02_parallax_3dof_vs_6dof.py
python -m http.server 8000
# open localhost:8000/dof_playground.html
```



→ Rotate the 3D window with the mouse; close it to continue.

→ Open outputs/dimensions.png and outputs/parallax.png.

→ In the playground press M and compare 3DoF and 6DoF.

Three things to remember

1. A dimension is a number you need; 2.5D depth keeps one value per line of sight, 3D keeps them all.
2. Useful 3D = geometry + topology + semantics. Point clouds are the raw geometry.
3. A 6DoF pose is 6 numbers. Only translation creates parallax, so 6DoF needs real 3D content.

BEFORE NEXT SESSION

Read the Ch. 2 maths section of the book. Lab L1 (with Day 2) is in the assignments document.